

Visual Causal-chain Bookmarking: Retrieval-Weighted Credit Assignment for Reinforcement Learning from Delayed Reward

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Abstract

Reinforcement learning (RL) agents trained with a single trajectory-level return face a structural credit-assignment problem: when a reward arrives many steps after the decision that caused it, every intermediate action receives the same learning signal, regardless of whether it was causally responsible for the outcome. This dissertation studies *Visual Causal-chain Bookmarking* (VCBM), a family of methods that concentrate credit on the specific decision points that determine an episode’s return rather than spreading it uniformly across the trajectory, and develops two domain-specific instantiations of this idea.

The first instantiation targets a fully observed, low-dimensional Markov decision process (a delayed-reward watch-repair task), where VCBM statistically discovers which state components are causally controlled by the agent’s actions and estimates per-decision-point action values directly; because the environment’s causal structure is verifiable, this instantiation is evaluated against known ground truth as well as against RUDDER [1], a return-redistribution baseline built on an LSTM contribution network. VCBM converges to a 95.07% correct-decision rate in a mean of $1,532.3 \pm 1,428.0$ episodes across 20 random seeds (range 1,196–7,757), against RUDDER’s slower and more variable convergence (mean $3,327.1 \pm 2,010.9$ episodes, median 2,829, range 1,603–10,936, computed from the 20 per-seed training logs), and its discovered causal structure exactly matches the environment’s true generative structure in all 20 seeds.

The second instantiation targets a partially observed, language-based setting – an LLM agent trained with Leave-One-Out Proximal Policy Optimisation (LOOP) [2] inside the AppWorld benchmark [3] – where causal discovery is intractable and a structural heuristic (state-mutating tool calls) substitutes for it, with credit distributed via cosine-similarity retrieval over an episodic memory of bookmarked turns and blended with LOOP’s trajectory-level advantage. TODO: state the headline quantitative result (e.g. convergence speed and/or final policy quality relative to the LOOP baseline) once the AppWorld LOOP-vs-VCBM CSF3 run completes; until then, this instantiation is a designed and implemented, but not yet empirically validated, contribution.

Taken together, the watch-repair result supports the dissertation’s central claim – that explicitly identifying and weighting the causally relevant steps of a trajectory is a more sample-efficient credit-assignment strategy than either an undifferentiated trajectory-level baseline or a learned return-redistribution network – in the one setting where it has so far been tested; TODO: extend this claim to the AppWorld setting once that comparison is complete.

Declaration of originality

I hereby confirm that no portion of the work referred to in the thesis has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Acknowledgments

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1 Introduction

Reinforcement learning (RL) agents that learn from a trajectory-level reward face a structural credit-assignment problem whenever that reward is *delayed* and *sparse*: it arrives only at the end of an episode, after dozens or hundreds of intermediate actions, most of which did not causally influence the outcome. This difficulty is not a corner case – it recurs across the trial-and-error paradigm’s main applications, from robotic control to the post-training alignment of large language models (LLMs) via reinforcement learning from human feedback (RLHF) [4], [5]. A watch-repair technician’s one decision to repair or not repair a watch determines the profit realised weeks later, after many unrelated logistics events; an LLM agent solving a multi-step task in a simulated software environment such as AppWorld [3] executes a long sequence of tool calls, of which only a handful actually change task-relevant state. In both cases, a policy-gradient update that assigns the same scalar advantage to every action in the trajectory is forced to average this delayed signal across many causally irrelevant steps, which slows convergence and can bias the policy towards whichever irrelevant actions happen to correlate with high-return trajectories.

Two established families of methods respond to this problem without questioning the underlying credit-assignment strategy. Trajectory-level baselines, such as the leave-one-out (LOO) advantage estimator used in LOOP [2] for training LLM agents, reduce the variance of a Monte-Carlo return by subtracting a baseline computed from other rollouts of the same scenario, but every token or action within a rollout still receives an identical, undifferentiated advantage. Return-redistribution methods such as RUDDER [1] instead train an auxiliary network – an LSTM that predicts the eventual return from every prefix of the trajectory – and redistribute reward as the difference between consecutive predictions, which in principle produces a denser and better-shaped reward signal. However, RUDDER’s redistribution is learned end-to-end from a recurrent model whose predictions are only as reliable as its own training data, so early in training, before the LSTM has converged, the redistributed reward is itself noisy, and the resulting variance can slow down the tabular or parametric policy that consumes it.

An underexploited alternative is to make credit assignment *structural* rather than purely statistical: identify which specific timesteps in a trajectory are causally responsible for the eventual outcome, and concentrate the learning signal on those timesteps directly, rather than learning a smooth redistribution over the whole sequence. This is the guiding idea behind *Visual Causal-chain Bookmarking* (VCBM), the method studied in this dissertation. VCBM is deliberately instantiated two different ways depending on what is tractable in the target environment. In a fully observed, low-dimensional Markov decision process (MDP), a genuine statistical test can determine which state components are causally controlled by the agent’s action, so VCBM performs explicit *causal discovery*: a two-proportion hy-

pothesis test classifies each state dimension as controlled, static, clock-like, or environmental noise, decision points (“bookmarks”) are identified as states matching a previously observed controlled-component change, and a Welford online estimator accumulates the return statistics of each bookmarked decision directly. In a partially observed, natural-language setting such as an LLM agent operating inside AppWorld, this statistical discovery procedure is intractable – the state is unstructured text, not a fixed set of interpretable components – so the second instantiation substitutes a structural heuristic (a turn is bookmarked if it issues a state-mutating tool call) for discovery, and replaces exact-match value lookup with retrieval: a lightweight text embedding of each bookmarked turn is stored in an episodic memory, and credit for a completed trajectory is distributed across bookmarks in proportion to their cosine similarity to the terminal state, blended with the trajectory-level LOO advantage that would otherwise be applied uniformly.

This dissertation’s technical contribution is the design, implementation, and empirical evaluation of both VCBM instantiations against their respective natural baselines. Section 2 formalises the shared MDP/RL notation and the two baseline algorithms – RUDDER’s LSTM-based return redistribution and LOOP’s leave-one-out PPO objective – that VCBM is compared against. Section 3 situates VCBM within the wider credit-assignment literature that motivates its design but is not directly benchmarked in Section 5, grouped into temporal/counterfactual redistribution methods, group-relative/trajectory-level RL fine-tuning methods, episodic-memory and retrieval mechanisms for LLM agents, and survey works establishing the problem’s scope. Section 4 presents the full mathematical formulation of both VCBM instantiations: the causal-discovery-and-Welford-estimation mechanism for the tabular watch-repair task, and the retrieval-and-blend mechanism for the AppWorld LLM-agent task, making explicit where the two instantiations share a common structure and where they necessarily diverge. Section 5 reports controlled multi-seed experiments in the watch-repair environment, where VCBM’s discovered causal structure is directly verifiable against the environment’s known ground truth, together with TODO: the corresponding AppWorld LOOP-vs-VCBM comparison once training completes. Section 6 concludes with a statement of technical achievement against the dissertation’s original objectives and discusses the limitations of the structural-heuristic instantiation as a direction for future work.

2 Preliminaries

2.1 Markov decision processes and policy-gradient reinforcement learning

An episodic MDP is a tuple $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $P(s_{t+1} \mid s_t, a_t)$, reward function $r_t = r(s_t, a_t)$, discount factor $\gamma \in (0, 1]$, and horizon T . A stochastic policy $\pi_\theta(a_t \mid s_t)$, parameterised by θ , generates a trajectory $\tau = (s_0, a_0, r_0, \dots, s_T)$ with realised

(undiscounted, as used throughout this dissertation’s environments) return

$$G(\tau) = \sum_{t=0}^T r_t. \quad (1)$$

Policy-gradient methods maximise $J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta}[G(\tau)]$ by ascending an estimate of $\nabla_\theta J(\theta) = \mathbb{E}_\tau[\sum_t \nabla_\theta \log \pi_\theta(a_t | s_t) A_t]$, where A_t is an *advantage* estimate that reduces gradient variance relative to using the raw return $G(\tau)$ directly. The two baselines studied in this dissertation, RUDDER and LOOP, both address the variance/credit-assignment problem inherent in estimating A_t from a sparse, trajectory-level reward, but with different mechanisms; the central hypothesis of this dissertation, tested via VCBM, is that a third mechanism – explicit identification of the causally relevant timesteps – offers a more sample-efficient path to a low-variance, well-shaped A_t than either.

2.2 RUDDER: LSTM-based return redistribution

RUDDER (RUDDER: Return Decomposition for Delayed Rewards) [1] addresses delayed, sparse reward by training an auxiliary sequence model to predict the eventual episode return from every prefix of the trajectory, then using the *change* in that prediction from one timestep to the next as a redistributed, per-step reward. Let $\Delta s_t = s_t - s_{t-1}$ denote the state-difference at timestep t . Given the trajectory’s state-difference and action sequence up to step t , an LSTM with disabled forget and output gates produces a return prediction at every prefix,

$$\hat{G}(s_{0:t}, a_{0:t}) = \text{Linear}(\text{LSTM}(\Delta s_{0:t}, a_{0:t}))_t, \quad (2)$$

trained against the true (scaled) episode return $G_0 = G(\tau)/R_{\text{scale}}$ with a combination of a terminal-step loss and an auxiliary loss applied at every prefix,

$$\mathcal{L}_{\text{RUDDER}} = \underbrace{(\hat{G}(s_{0:T}) - G_0)^2}_{\mathcal{L}_{\text{main}}} + \lambda_{\text{aux}} \underbrace{\frac{1}{T} \sum_{t=0}^T (\hat{G}(s_{0:t}) - G_0)^2}_{\mathcal{L}_{\text{aux}}}. \quad (3)$$

The redistributed reward at each timestep is then the scaled first difference of consecutive predictions,

$$\tilde{r}_t = R_{\text{scale}} \cdot (\hat{G}(s_{0:t}) - \hat{G}(s_{0:t-1})), \quad \hat{G}(s_{0:-1}) \equiv 0, \quad (4)$$

which sums telescopically to the true episode return, $\sum_t \tilde{r}_t = R_{\text{scale}} \cdot \hat{G}(s_{0:T}) \approx G(\tau)$, while ideally concentrating non-zero reward at the timesteps the LSTM has learned are predictive of the outcome. Because $\tilde{r}_t$ is only as accurate as the LSTM’s own convergence, RUDDER’s redistribution is a *learned, statistical* solution to credit assignment: it does not identify causal structure directly, but approximates it

through gradient-based function approximation, and is used in this dissertation as the representative baseline against which VCBM’s structural approach is compared in a tabular setting (Section 5).

2.3 LOOP: leave-one-out advantage estimation with PPO-clipped updates

LOOP addresses the same delayed-reward problem in a different regime – training an LLM policy via multi-turn tool-use rollouts, where a value-function critic is memory-expensive to maintain alongside a large language model. Given K rollouts $\tau^{(1)}, \dots, \tau^{(K)}$ sampled from the *same* task scenario, each with realised return $G^{(k)} = G(\tau^{(k)})$, the leave-one-out baseline for rollout k is the mean return of the *other* $K - 1$ rollouts,

$$b^{(k)} = \frac{1}{K-1} \sum_{j \neq k} G^{(j)}, \quad A^{(k)} = G^{(k)} - b^{(k)}, \quad (5)$$

which is applied uniformly to every token generated within rollout k – i.e. $A_t = A^{(k)}$ for all t in that rollout, the trajectory-level credit-assignment limitation that VCBM’s blend mechanism (Section 4) directly targets.

This uniform advantage is then consumed by a PPO-clipped policy-gradient objective, which is needed because rollouts are generated by a separate, possibly numerically divergent inference engine. The policy-gradient update requires an importance weight correcting for the mismatch between the log-probability under which a token was sampled, $\pi_{\theta_{\text{old}}}$, and the log-probability under the current training weights, π_{θ} : $w_t = \exp(\log \pi_{\theta}(a_t) - \log \pi_{\theta_{\text{old}}}(a_t))$. The resulting per-token objective applies a PPO-style clip [2] to this importance weight to bound the size of any single update,

$$\mathcal{L}_{\text{PPO}}(t) = -\min(w_t A_t, \text{clip}(w_t, 1 - \epsilon, 1 + \epsilon) A_t), \quad (6)$$

with clip range ϵ (default 0.1 in the AppWorld configuration used in this dissertation). Equation (6) is the loss into which VCBM’s retrieval-weighted, per-token advantage (Section 4.2) is substituted in place of the uniform $A_t = A^{(k)}$, so that VCBM modifies *which* advantage each token receives, without requiring any change to the surrounding PPO-clipped policy-gradient machinery.

3 Related work

Section 2 introduced RUDDER and LOOP because VCBM is directly benchmarked against them. This section situates VCBM against the wider credit-assignment literature that motivates its design but is not directly benchmarked in Section 5, organised into four thematic groups: temporal/counterfactual credit-assignment methods beyond RUDDER; group-relative and trajectory-level RL

fine-tuning methods beyond LOOP; episodic memory and retrieval mechanisms for long-horizon LLM agents; and survey works that establish the scope of the credit-assignment problem VCBM addresses.

3.1 Temporal and counterfactual credit assignment beyond RUDDER

RUDDER (Section 2.2) is one member of a broader family of methods that redistribute a sparse terminal reward across a trajectory by learning an auxiliary predictive model, rather than by testing for causal structure directly. Hindsight Credit Assignment [6] assigns credit to past actions using a hindsight distribution conditioned on the outcome actually observed, which – like RUDDER’s LSTM – is a *learned, statistical* redistribution rather than a discovered causal one; VCBM’s tabular instantiation differs by testing directly, via the two-proportion z -test of Equation (8), which state components an action can influence, rather than fitting a conditional distribution over outcomes. Counterfactual Credit Assignment [7] and COCOA [8] go further towards VCBM’s causal framing: both use structural causal models or model-based counterfactual queries to estimate counterfactual returns, which is conceptually closer to VCBM’s causal-role classifier than RUDDER’s or Hindsight Credit Assignment’s purely predictive approach. The distinction that remains is that Counterfactual Credit Assignment and COCOA both require a learned forward or counterfactual model of the environment to answer their counterfactual queries, whereas VCBM’s tabular causal-role test (Equation (7)) requires no model at all – only online frequency counts – which trades the generality of a learned counterfactual model for a test that is exact and directly verifiable against ground truth, as demonstrated in Section 5.1. Randomized Return Decomposition [9] and ϕ -PPO’s Shapley-value credit assignment [10] represent two further redistribution strategies – respectively a randomised decomposition objective and a game-theoretic (Shapley) attribution of the return across timesteps – both of which, like RUDDER, redistribute credit continuously over every timestep rather than concentrating it on a discrete, discovered set of bookmarks; VCBM’s central departure from this whole family is that a timestep either is or is not a bookmark, with no residual credit paid to timesteps in between.

3.2 Group-relative and trajectory-level RL fine-tuning beyond LOOP

LOOP’s leave-one-out advantage (Section 2.3) is one instance of a broader class of methods that compute a single scalar advantage per trajectory and apply it uniformly to every token or turn within that trajectory. Multi-turn GRPO variants [11] extend group-relative advantage estimation from the single-response setting to multi-turn reasoning, but the description available for this line of work indicates the resulting advantage is still assigned per turn from a group-relative baseline rather than per bookmark from a retrieval-weighted memory, so it addresses coarser temporal granularity than

VCBM’s turn-level, similarity-weighted credit (Equation (11)). G2PO [12] is a closer point of comparison because it is evaluated on AppWorld directly: it builds a global state-transition graph to improve credit allocation across long-horizon benchmarks including AppWorld, which – like VCBM’s structural bookmark rule (Section 4.2) – exploits the state-transition structure of the environment rather than the reward alone. The distinction is one of mechanism: G2PO’s credit allocation is read off a learned or constructed graph over the whole state-transition history, whereas VCBM’s AppWorld instantiation uses a much simpler structural heuristic (a turn is a bookmark if and only if it issues a state-mutating tool call) combined with a flat embedding-similarity retrieval, trading G2PO’s richer global structure for a mechanism that is cheap enough to run inline during PPO-clipped training with no additional network calls. SPA-RL [13] densifies rewards along a reasoning path via stepwise progress attribution, which – similarly to VCBM – produces a non-uniform, step-level reward signal from a single trajectory-level outcome, but does so by attributing *progress* at every step rather than selecting a discrete subset of causally relevant steps; VCBM’s bookmark mechanism is, by construction, sparse where SPA-RL’s progress attribution is dense.

3.3 Episodic memory and retrieval in long-horizon LLM agents

The retrieval mechanism in VCBM’s AppWorld instantiation (Equation (11)) draws on a longer tradition of using an external memory of past experience to inform credit assignment or value estimation. Memory-R2 [14] is the closest single point of comparison in the surveyed literature: it addresses fair credit assignment for long-horizon, memory-augmented LLM agents via a method called LoGo-GRPO combined with curriculum training, and its use of a memory structure to shape per-turn or per-token credit is architecturally close to VCBM’s own blend of a bookmark memory with a trajectory-level advantage (Equation (12)). Based on the one-sentence description available, the distinction that can honestly be drawn is that LoGo-GRPO folds its memory-based credit into a group-relative policy-optimisation objective (GRPO) together with a curriculum-training schedule, whereas VCBM keeps the memory-weighting step (retrieval and softmax weighting over bookmarked turns) and the underlying policy-gradient objective (PPO-clipped LOOP) as two separable stages connected only by the mixing coefficient β in Equation (12) – but the available description does not specify LoGo-GRPO’s retrieval mechanism precisely enough to state with confidence that this separation is a substantive rather than superficial difference. This dissertation treats that as an open question rather than a claimed point of novelty; the possibility that VCBM’s AppWorld mechanism is a special or simplified case of LoGo-GRPO’s memory-augmented credit assignment is a limitation worth carrying into Section 6. A generally curious agent [15] instead uses curiosity signals, rather than retrieval over past bookmarked outcomes, to guide long-horizon exploration and credit assignment, which addresses a different sub-problem (what to explore) from the one VCBM addresses (which already-

taken actions to credit); the two are complementary rather than competing mechanisms.

3.4 Surveys establishing the scope of the credit-assignment problem

Three survey-style works help frame where VCBM sits within the credit-assignment literature as a whole. Pignatelli et al. [16] survey temporal credit assignment in deep RL broadly, providing a unifying formalism and evaluation protocols that this dissertation’s tabular experiment (Section 5.1) is consistent with in spirit, if not evaluated against directly. Two further surveys narrow this scope specifically to LLM agents: an agentic RL survey [17] identifies credit assignment as a central bottleneck in scaling LLM agents, and a companion survey [18] organises the resulting methods by granularity (token-, turn-, or trajectory-level) and methodology, emphasising long-horizon challenges. A survey of long-horizon agents [19] reviews hierarchical and credit-assignment techniques spanning both manipulation and LLM-agent tasks. Read together, these surveys indicate that the field has largely converged on *learned* solutions – graph-based, curriculum-based, or attention/attribution-based – to the granularity problem that VCBM instead addresses, in its tabular instantiation, with an explicit statistical test.

3.5 VCBM’s distinguishing position

Across all three groups above, the great majority of surveyed methods – Hindsight Credit Assignment, Randomized Return Decomposition, ϕ -PPO, Multi-turn GRPO, G2PO, SPA-RL, and the memory-augmented and curiosity-driven agent methods alike – assign credit through a *learned or continuously-valued* mechanism: a fitted hindsight distribution, a randomised decomposition objective, a Shapley-value estimator, a group-relative baseline, a constructed graph, a progress-attribution model, or a curiosity signal. VCBM’s two instantiations are deliberately harder-edged than this. In the tabular case (Section 4.1), a bookmark either is or is not causally controlled, decided by a hypothesis test with a fixed significance threshold, and credit is not smoothed across non-bookmarked timesteps at all – they are excluded from the value-estimation key entirely. In the AppWorld case (Section 4.2), the decision of what counts as a bookmark is similarly hard – a structural rule (state-mutating tool call), not a learned classifier – although the *weighting* of retrieved bookmarks is soft (a temperature-scaled softmax over cosine similarity, Equation (11)), which is where VCBM’s AppWorld instantiation is closest to, and hardest to cleanly distinguish from, memory-augmented methods such as Memory-R2’s LoGo-GRPO (Section 3.3). The counterfactual model-based methods (Counterfactual Credit Assignment, COCOA) are the exception that shares VCBM’s causal framing without sharing its hardness, since they substitute a learned counterfactual model for VCBM’s model-free statistical test. This dissertation’s contribution is therefore best read not as introducing causal or memory-based

credit assignment as a category – both are well established in the literature surveyed above – but as testing, in two contrasting regimes, how far a comparatively hard, discovery-or-heuristic-driven selection of bookmarks can go before the softer, learned mechanisms that dominate the surveyed literature become necessary.

4 Technical contribution

Both instantiations of VCBM developed in this dissertation share a common three-stage template – *detect* the decision points that plausibly carry causal weight, *store* them in a structured memory, and *weight* the learning signal according to that memory – but the concrete mechanism at each stage necessarily differs between a fully observed tabular MDP, where causal discovery is statistically tractable, and a partially observed, language-based MDP, where it is not. This section presents both instantiations in full, and states explicitly, rather than implicitly, where the two diverge.

4.1 VCBM for tabular MDPs: causal discovery and Welford estimation

4.1.1 Motivation

In a fully observed MDP with a small, fixed-dimensional state space, it is possible to test directly, from interaction data alone, which state components an agent’s action can causally influence and which change independently of it. If such a test is available, credit for the final return can be assigned exactly to the decision points where a causally controlled change originates, rather than diffused across an entire trajectory of otherwise uninformative filler steps – as occurs, for instance, in the watch-repair environment of Section 5, where a single decision at $t = 0$ determines the return realised 50 steps later.

4.1.2 Causal-role discovery

Let the state be decomposed into n scalar components $s = (s^{(1)}, \dots, s^{(n)})$. For each component i and each action value $a \in \{0, 1\}$, a running count $c_{i,a}$ tracks how often component i changed value on a step where action a was taken, out of n_a total steps with that action. After a warm-up period, each component is classified into one of four causal roles using its empirical change rate

$\rho_{i,a} = c_{i,a}/n_a$:

$$\text{role}(i) = \begin{cases} \text{CLOCK} & \min(\rho_{i,0}, \rho_{i,1}) > \eta_{\text{clock}} \\ \text{STATIC} & c_{i,0} + c_{i,1} = 0 \\ \text{CONTROLLED} & z_i > z_{\text{thresh}} \\ \text{NOISE} & \text{otherwise,} \end{cases} \quad (7)$$

where z_i is the two-proportion test statistic comparing the change rate under each action,

$$z_i = \frac{|\rho_{i,0} - \rho_{i,1}|}{\sqrt{\hat{p}(1 - \hat{p})(1/n_0 + 1/n_1)}}, \quad \hat{p} = \frac{c_{i,0} + c_{i,1}}{n_0 + n_1}. \quad (8)$$

A component is CONTROLLED only if it changes at a rate that depends significantly on the action taken; CLOCK and STATIC components change (almost) deterministically regardless of action and can never carry action-dependent credit; NOISE components change at a rate that does not depend significantly on the action, given the available sample size, and are excluded from the value-estimation key described below. This test is what allowed VCBM, without any hand-specified labels, to recover the causal structure of the watch-repair environment reported in Section 5.

4.1.3 Bookmark discovery and value estimation

A state is flagged as a *bookmark opportunity* – a decision point worth tracking – if its projection onto the CONTROLLED and CLOCK components matches a context previously observed immediately before a CONTROLLED component changed value. At each bookmark, a compact key is formed by projecting the state onto its STATIC, CONTROLLED, and CLOCK components only (marginalising out NOISE, which by construction cannot carry action-relevant signal), and a Welford online estimator accumulates, per (key, action) pair, the mean and variance of the *full* episode return $G(\tau)$ observed whenever that bookmark/action pair occurred:

$$\mu_n = \mu_{n-1} + \frac{G(\tau) - \mu_{n-1}}{n}, \quad M_{2,n} = M_{2,n-1} + (G(\tau) - \mu_{n-1})(G(\tau) - \mu_n), \quad (9)$$

with standard error $\text{SE} = \sqrt{M_{2,n}/(n(n-1))}$. Because the key marginalises out noise components, samples from many episodes that differ only in their noise trajectory are correctly pooled into the same value estimate – a form of sample sharing unavailable to a plain per-state value table.

4.1.4 Action selection

At a bookmark, actions are sampled uniformly until both actions have accumulated at least $n_{\min}$ visits; thereafter, a two-sample z -test on the two actions' Welford means, $z = |\mu_0 - \mu_1| / \sqrt{\text{SE}_0^2 + \text{SE}_1^2}$, determines whether the value difference is statistically significant. If significant, the higher-mean action is chosen greedily (with a small residual exploration probability $\epsilon_{\min}$); otherwise, an ϵ -greedy rule with decaying ϵ continues exploring. At non-bookmark timesteps, an action is drawn uniformly at random, since Equation (7) guarantees such steps do not causally affect the outcome once the classifier has converged.

4.2 VCBM for language-based MDPs: retrieval-weighted credit blending

4.2.1 Motivation

The causal-role test of Section 4.1 requires a fixed, low-dimensional, fully observed state representation whose components can be tracked and change-tested individually. An LLM agent operating in AppWorld [3] instead observes unstructured natural-language tool outputs, with no fixed component decomposition and no tractable way to run a per-component hypothesis test online during training. The second VCBM instantiation therefore replaces statistical causal *discovery* with a structural *heuristic* for what counts as a bookmark, and replaces exact-match value lookup with similarity-based *retrieval*, since no two AppWorld episodes share an identical state trajectory the way watch-repair episodes do.

4.2.2 Bookmark detection and episodic memory

A trajectory turn is bookmarked if it issues a state-mutating tool call (as opposed to a read-only or no-op action), detected directly via the environment server's execution interface with no additional network calls. Each bookmarked turn's tool-output text x is embedded with a deterministic, dependency-free hashing-trick bag-of-words encoding g_ϕ into a fixed-dimension vector,

$$g_\phi(x)_j = \frac{1}{\|v\|_2} \sum_{\text{token} \in x} \mathbb{1}[h(\text{token}) = j], \quad h(\text{token}) = \text{int}(\text{MD5}(\text{token})[:8]) \bmod d, \quad (10)$$

and stored in an episodic memory bank as a tuple $(\text{turn_idx}, g_\phi(x), G(\tau))$ – the bookmark's position, its text embedding, and the return of the trajectory it belonged to.

4.2.3 Retrieval and temperature-scaled weighting

Given a completed trajectory’s terminal-observation embedding $z_H = g_\phi(x_H)$, credit is distributed across the top- k most similar bookmarks retrieved from memory by cosine similarity,

$$\text{sim}(z_H, k_i) = \frac{z_H \cdot k_i}{\|z_H\| \|k_i\|}, \quad w_i = \frac{\exp(\text{sim}_i/\tau)}{\sum_{j \in \text{top-}k} \exp(\text{sim}_j/\tau)}, \quad (11)$$

with temperature τ controlling how sharply weight concentrates on the single most similar retrieved bookmark. Tokens belonging to a retrieved bookmarked turn receive weight w_i ; all other tokens receive a fixed floor weight $\alpha \in (0, 1)$ so that non-bookmarked turns are down-weighted but never entirely excluded from the gradient; the resulting per-token weight vector is renormalised to sum to the number of output tokens, preserving the overall gradient scale of the underlying loss.

4.2.4 Blending with the trajectory-level LOO advantage

Rather than replacing LOOP’s leave-one-out advantage outright, the per-token VCBM weight $\tilde{w}_t$ (Equation (11), renormalised) is combined with the uniform LOO advantage $A^{(k)}$ (Equation (5)) via a mixing coefficient $\beta \in [0, 1]$:

$$A_{\text{VCBM},t} = A^{(k)} \cdot \tilde{w}_t, \quad A_{\text{blend},t} = (1 - \beta) A^{(k)} + \beta A_{\text{VCBM},t}, \quad (12)$$

which reduces exactly to plain LOOP at $\beta = 0$ and to a fully bookmark-concentrated advantage at $\beta = 1$; $A_{\text{blend},t}$ is substituted directly for the uniform A_t in Equation (6). Rollouts containing zero bookmarked turns receive the unmodified scalar LOO advantage, since Equation (12) is undefined without at least one retrieval candidate.

4.3 Where the two instantiations agree and where they necessarily diverge

Both instantiations follow the detect/store/weight template and are motivated by the same hypothesis – that concentrating credit on causally relevant decision points outperforms an undifferentiated trajectory-level signal – but they are not the same algorithm, and this dissertation does not claim they are. Table 1 makes the divergence explicit: the tabular instantiation *discovers* its bookmarks through a statistical hypothesis test with no learned embedding or similarity search anywhere in the pipeline, whereas the AppWorld instantiation *assumes* a structural rule for what constitutes a bookmark and relies on embedding-space retrieval, absent from the tabular version, to generalise across episodes that never share an identical state. The two instantiations are best understood as two different answers to the same design question – “how should credit be concentrated when discovery is/is not

statistically tractable” – evaluated independently against the strongest available baseline in each regime, rather than as one method validated twice.

Table 1. Mechanistic comparison of the two VCBM instantiations studied in this dissertation.

Aspect	Tabular (Section 4.1)	AppWorld/LLM (Section 4.2)
Bookmark detection	Statistical causal-role test (Eq. (7))	Structural rule (state-mutating tool call)
Memory content	Welford (μ, σ^2) per (key, action)	Raw (index, embedding, return) tuples
Retrieval mechanism	Exact key match	Cosine-similarity top- k (Eq. (11))
Underlying policy	Tabular Q -style controller	LoRA-adapted LLM, PPO-clipped (Eq. (6))
Fallback when data is scarce	Balanced random sampling	Flat floor weight α

4.4 Six structural capabilities VCBM has that a uniform trajectory-level baseline lacks

The detect/store/weight template formalised in Sections 4.1 and 4.2 gives both VCBM instantiations six mechanistic capabilities that LOOP’s leave-one-out advantage (Equation (5)) structurally cannot have, because that advantage is a single scalar $A^{(k)}$ applied uniformly to every token in rollout k , with no notion of *which* token mattered. First, VCBM performs *fine-grained credit assignment*: rather than one signal for the whole trajectory, the causal-role test (Equation (7)) in the tabular case and the bookmark rule in the AppWorld case each identify specific timesteps as the locus of credit, so the advantage a token receives depends on whether that token is a bookmark at all. Second, VCBM produces *per-decision value estimates* rather than a per-rollout average: the Welford estimator of Equation (9) maintains a mean and variance keyed to an individual (bookmark, action) pair, and the retrieval-softmax of Equation (11) likewise assigns weight w_i to an individual retrieved turn, not to the rollout as a whole. Third, only the tabular instantiation has genuine *causal/structural knowledge of the environment*: Equation (7) explicitly separates CONTROLLED components, which the agent’s action can influence, from CLOCK, STATIC, and NOISE components, which change independently of it; the AppWorld instantiation substitutes a structural heuristic for this discovery step (Section 4.2) and so has this capability only in a weaker, assumed form. Fourth, both instantiations maintain a *memory of past experiences* that persists beyond a single batch of rollouts – the Welford accumulator’s running (key, action) statistics and the AppWorld episodic memory bank of $(\text{turn_idx}, g_\phi(x), G(\tau))$ tuples both carry information forward across training iterations, unlike LOOP’s leave-one-out baseline, which is recomputed from scratch within each group of K rollouts and discarded once that group’s update is applied. Fifth,

the AppWorld instantiation specifically adds *retrieval-based generalisation*: Equation (11) matches a new trajectory’s terminal state to similar past bookmarks by cosine similarity and routes credit accordingly, rather than judging a rollout only against the $K - 1$ other rollouts sampled for the same scenario in the current batch. Sixth, both instantiations perform *selective, weighted credit assignment*: the tabular value-estimation key marginalises out NOISE components entirely (Section 4.1), and the AppWorld blend (Equation (12)) concentrates weight $\tilde{w}_t$ on retrieved bookmarks while down-weighting, but not eliminating, all other tokens via the floor weight α – in both cases the learning signal is deliberately non-uniform, in contrast to LOOP’s single $A^{(k)}$ applied identically to every token.

Each of these six structural capabilities carries a plausible practical advantage, reasoned directly from the mechanism just described, but the evidentiary status of these advantages differs sharply between the two instantiations and must not be overstated. Fine-grained credit assignment is expected to lower gradient variance, because irrelevant steps no longer dilute a signal that a single relevant step should dominate, and to reduce the risk of reinforcing spurious action-outcome correlations that a uniform advantage cannot distinguish from genuine ones. Per-decision value estimates are expected to enable targeted improvement at the exact point of failure, rather than only the coarser signal that “this whole episode was bad”, which should be especially valuable in tasks with one or a few pivotal decisions, such as the single $t = 0$ repair decision in the watch-repair environment. Verified causal knowledge is expected to prevent wasted exploration on components the agent does not control once the classifier has converged (Section 4.1 already exploits this at non-bookmark timesteps), to generalise more robustly than a purely statistical trajectory-outcome association, and – uniquely among the six, because Equation (7)’s output is a discrete label per component – to be independently verifiable against ground truth, exactly as demonstrated in Section 5.1. Persistent episodic memory is expected to avoid re-learning the same lesson from scratch in every batch and to pool evidence across episodes that differ only in causally irrelevant details faster than treating every episode as fully independent. Retrieval-based generalisation is expected to apply lessons from structurally similar past situations to new ones not seen in the current batch of K rollouts, which should matter most in long-horizon, highly varied task distributions such as AppWorld, where exact scenario repeats are rare. Selective, weighted credit is expected to concentrate learning capacity where it has real leverage over the outcome and to improve training stability by reducing the chance the policy drifts on tokens or actions that merely co-occurred with a high-return rollout rather than causing it.

These are, at this stage, the *design rationale* for why VCBM’s detect/store/weight structure was built this way, not a set of uniformly confirmed empirical results. Of the practical advantages above, only those attributable to fine-grained credit assignment, per-decision value estimates, and verifiable causal knowledge have been empirically observed, and only in the single setting reported in Section 5.1: VCBM’s lower-mean, lower-variance, lower-maximum episodes-to-converge distribution

relative to RUDDER (Table 2) is consistent with reduced gradient variance and faster convergence from concentrating the signal on one causally verified decision point, and the exact recovery of the environment’s true causal structure in all 20 seeds is the verifiability advantage directly demonstrated. Even that comparison remains qualitative pending the paired significance test flagged in Section 5.1. The remaining three advantages – persistent memory avoiding batch-to-batch re-learning, retrieval-based generalisation across structurally similar but non-identical episodes, and selective weighting improving training stability – are structurally present in the AppWorld instantiation by construction (Section 4.2) but have not yet been tested empirically at all: they remain hypotheses attached to a mechanism that has been implemented but not yet evaluated against plain LOOP, pending the AppWorld comparison flagged as TODO in Section 5.2. This dissertation therefore does not claim that VCBM outperforms LOOP in the AppWorld setting; it claims only that VCBM’s mechanism structurally possesses these six capabilities, that the subset of associated advantages testable in the tabular setting was observed there, and that the corresponding claim for the language-agent setting awaits the pending run.

5 Experiments

5.1 Controlled evaluation: RUDDER vs. VCBM in a delayed-reward tabular MDP

5.1.1 Environment

The watch-repair environment models a single consequential decision under delayed, stochastic feedback. At $t = 0$ the agent observes a randomly drawn watch BRAND $b \sim \text{Uniform}\{0, 1, 2, 3\}$ and chooses whether to repair it ($a = 0$, incurring an immediate stochastic cost $r^{\text{repair}} \sim \mathcal{N}(\mu_{\text{repair}}[b], \sigma_{\text{repair}}^2[b])$) or not ($a = 1$, no cost). For the remaining 50 timesteps, two independent NOISE counters increment stochastically at fixed, action-independent rates, and a deterministic CLOCK component advances every step; the agent’s action has no further effect on the environment after $t = 0$. Only if the watch was repaired does the episode yield a terminal reward at $t = 50$, equal to the brand’s resale price minus accumulated transport-condition costs and a stochastic transport penalty. Because prices, repair costs, and transport-cost sensitivities differ by brand, the return-maximising policy is brand-dependent (repairing is optimal only for brands 1 and 2), giving a known ground-truth optimal action per state that is used purely for evaluation, never as a training signal.

5.1.2 Method and metrics

Both RUDDER (Section 2.2, redistributing reward into a tabular direct- Q -estimation update) and VCBM (Section 4.1) were trained for 20 independent random seeds each, with training terminated once the moving average (750-episode window) fraction of optimal first-decisions reached the 95% target. Table 2 reports episodes-to-convergence, our primary sample-efficiency metric, alongside the final converged decision quality.

Table 2. Watch-repair environment: episodes required to reach 95% correct first-decisions, 20 seeds each. RUDDER statistics computed from the 20 individual per-seed training logs (Section 5.1); VCBM statistics taken from the method’s own multi-seed summary log.

Method	Seeds converged	Episodes to converge (range) ↓	Episodes to converge (mean $\pm$ std) ↓	Final good-decision % ↑
RUDDER	20/20	1,603 – 10,936	3,327.1 $\pm$ 2,010.9 (median 2,829)	95.07% (by construction of stopping rule)
VCBM	20/20	1,196 – 7,757	1,532.3 $\pm$ 1,428.0 (median 1,204.5)	95.07 $\pm$ 0.00%

5.1.3 Discovered causal structure

Across all 20 VCBM seeds, the causal-role classifier (Equation (7)) recovered an identical label assignment – CONTROLLED, NOISE, NOISE, STATIC, CLOCK – for the environment’s five state components (repaired-flag, two transport-condition counters, brand, timestep), exactly matching the environment’s true generative structure described above: the repaired-flag is the only component whose change rate depends on the agent’s action, the two transport counters change at fixed action-independent rates, the brand is fixed for the episode, and the timestep advances deterministically. This is direct evidence that VCBM’s discovery mechanism (Section 4.1) recovers ground-truth causal structure from interaction data alone, without being told in advance which of the five components matters.

5.1.4 Interpretation

Both methods reach the 95% target in all 20 seeds, but VCBM’s episodes-to-converge distribution has a lower minimum (1,196 vs. 1,603) and a lower maximum (7,757 vs. 10,936) than RUDDER’s, consistent with the hypothesis that concentrating the learning signal on the single causally relevant

decision point converges at least as fast as, and with less tail risk than, learning a smooth LSTM-based redistribution over the full 50-step trajectory. TODO: this interpretation is currently qualitative; add a paired significance test (e.g. Mann–Whitney U, appropriate given the right-skewed, non-normal distributions evident from the mean/median gap in Table 2) to substantiate the comparison once RUDDER’s full 20-seed distribution is available in the same structured form as VCBM’s summary log.

5.2 Large-scale evaluation: LOOP vs. VCBM-blended LOOP on AppWorld

TODO: this subsection reports the AppWorld comparison between the plain LOOP baseline (Section 2.3, Qwen-2.5-7B-Instruct policy, LoRA-adapted, FSDP2 training, $\beta = 0$ in Equation (12)) and VCBM-blended LOOP ($\beta = 1$) on identical GPU allocation, task split, and hyperparameters, once the corresponding CSF3 training runs complete. At minimum this subsection should report, per method: mean episode return over training iterations (learning curve), final held-out evaluation success rate (`r1.eval` split, evaluated every 5 iterations per the run configuration), and wall-clock/sample efficiency to reach a fixed return threshold, mirroring the metrics reported for the tabular experiment in Section 5.1 so the two experiments are directly comparable in structure even though their absolute metrics differ.

6 Conclusions

This dissertation set out to test whether reinforcement learning credit assignment can be improved by explicitly identifying the causally relevant decision points in a trajectory, rather than relying solely on trajectory-level baselines (as in LOOP) or learned return redistribution (as in RUDDER). Two domain-appropriate instantiations of this idea, jointly termed Visual Causal-chain Bookmarking (VCBM), were designed, implemented, and evaluated: a statistical causal-discovery-and-value-estimation mechanism for a fully observed tabular MDP, and a structural-heuristic-plus-retrieval mechanism for a partially observed LLM-agent setting.

In the tabular watch-repair environment, VCBM recovered the environment’s true causal structure from interaction data alone across all 20 evaluated seeds and converged to the target decision quality with a narrower and lower episode-count distribution than the RUDDER baseline (Section 5.1), directly supporting the dissertation’s central hypothesis in a setting where ground truth is fully verifiable. This evidence is currently confined to that one setting: TODO: once the AppWorld LOOP-vs-VCBM comparison (Section 5.2) completes, this paragraph must state explicitly whether the retrieval-weighted blend produced a measurable improvement over plain LOOP, since Technical Achievement is assessed primarily on demonstrated, not merely designed, contribution.

The dissertation is explicit, rather than silent, about a limitation of the AppWorld instantiation: unlike the tabular version, its bookmark detector is a hand-specified structural rule (state-mutating tool calls), not a learned or statistically discovered criterion, so it cannot be assumed to generalise to task domains where the causally relevant actions are not synonymous with state-mutating calls – a text-only reasoning task with no external tool-use, for instance, would have no bookmarks under the current rule at all. A natural direction for future work is a learned bookmark detector for the language setting, trained to predict causal relevance from the same kind of interaction data the tabular classifier already exploits (Equation (7)), which would close the conceptual gap between the two instantiations identified explicitly in Table 1 and bring the language-agent instantiation closer to a genuine causal-discovery mechanism rather than a structural proxy for one.

A second limitation, raised in Section 3.3, concerns novelty rather than performance: Memory-R2’s LoGo-GRPO [14] also uses a memory structure to shape per-turn credit for long-horizon, memory-augmented LLM agents, and the one-sentence description available for that method does not give enough detail to state confidently that VCBM’s separable retrieval-then-blend mechanism (Equation (12)) is mechanistically distinct from it, rather than a simplified special case. This dissertation has not verified that distinction against the LoGo-GRPO paper itself, and does not claim the AppWorld instantiation of VCBM is unambiguously novel relative to it; establishing (or refuting) a substantive difference is left as a direction for future work, alongside the pending AppWorld empirical comparison to LOOP itself.

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Appendices

Only add the project outline and risk assessments as defined in the first deliverable for this unit.